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MILITARY SPECIFICATION

CABLE ASSEMBLY, TORPEDO CONTROL, MARK 6 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for the procurement of cutterless Torpedo Control Cable Mk 6 Mod 0, LD 162420. The cable is used with deep depth torpedoes.

2. APPLICABLE DOCUMENTS

2.1 Other publications. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation - Packaging, Methods of
MIL-S-901	Shock Tests, H. I. (High Impact); Shipboard Machinery, Equipment and Systems; Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1355

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MIL-T-18404	Torpedoes, Environmental Requirements; General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-889	Dissimilar Metals

DRAWINGS

Naval Sea Systems Command (Code Ident 53711)

LD 162420	Cable, Torpedo Control, Mk 6 Mod 0
LD 162422	Shipping Condition, Cable, Torpedo Control; Mk 6 Mod 0
LD 162596	Torpedo Connector Receptacle Assembly
LD 162838	Connector, Torpedo Tube Cable, Mk 42 Mod 1
PL 2865969	Torpedo Connector Receptacle Assembly
DWG 758008	Torpedo Connector Receptacle Assembly
DWG 783552	Cable, Torpedo Control, Mk 6 Mod 0
DWG 783585	Screw, Shear
DWG 821024	Test Notes
DWG 2865440	Connector, Torpedo Tube Cable, Mk 42 Mod 1

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Description. Torpedo Control Cable Mk 6 Mod 0 is a cutterless, reuseable A-Cable used to transmit fire control signals to Torpedo Mk 48 and Mobile Target Mk 27.

3.2 Classification of requirements. The requirements for the Torpedo Control Cable are classified herein as follows:

Preproduction sample	3.3
Material	3.4
Design and construction	3.5
Mechanical performance	3.6
Electrical performance	3.7
Environmental	3.8
Workmanship	3.9

3.3 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample of four control cable assemblies shall be manufactured using the methods and procedures proposed for the production lot. These units are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing cables that comply with the specified requirements herein and in the contract. See 6.2.

3.4 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the control cable assemblies shall be in accordance with the requirements specified on the drawings and specifications in LD 162420. See 6.2.

3.4.1 Dissimilar metals. When employed in intimate contact with each other, dissimilar metals, as defined in MIL-STD-889, shall be suitably protected against electrolytic corrosion.

3.5 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the control cable assemblies shall be in accordance with the applicable drawings, specifications, and publications listed in LD 162420.

3.5.1 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 162420.

3.6 Mechanical performance requirements.

3.6.1 Torpedo cable assembly pull test. The cable assembly shall withstand a tensile force of 300 pounds (1.33 kilonewtons (kN)) (See 6.3) for a period of not less than 30 seconds without loss of conductor continuity when tested in accordance with 4.6.2.

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3.6.2 Connector mating and unmating forces. The torpedo tube cable connector, LD 162838, shall fully engage or disengage with an axial force of not greater than 50 pounds (224N). The cable assembly shall separate from the Torpedo Connector Receptacle, LD 162596, with an axial force of not more than 300 pounds (1.33kN) when tested in accordance with 4.6.3.

3.7 Electrical performance requirements.

3.7.1 Current carrying capacity. Circuits F, AA, and AB shall be capable of carrying 5.5 amperes (A) at 400 Hertz (Hz) for 20 minutes followed by a surge to 9.2 A for 3 seconds. In addition, any three other circuits selected at random shall be capable of carrying 5.5 A at 60 Hz for 20 minutes followed by a surge to 9.2 A for 3 seconds when tested in accordance with 4.6.4.

3.7.2 Dielectric strength. The assembly shall show no evidence of flash-over or dielectric breakdown when 725 ± 25 volts dc or 515 ± 15 volts ac is applied to any conductor with all other conductors at ground potential when tested in accordance with 4.6.5.

3.7.3 Insulation resistance. The insulation resistance of any cable conductor shall be greater than 10 megohms when 500 to 750 Vdc is applied to that conductor with all other conductors at ground potential when tested in accordance with 4.6.6.

3.7.4 Circuit resistance. The resistance of each circuit shall not exceed 0.50 ohm (Ω) with the cable mated to the Torpedo Connector Receptacle, LD 162596, and tested in accordance with 4.6.7.

3.7.5 Cable continuity. All conductors shall evidence electrical continuity when tested in accordance with 4.6.8.

3.8 Environmental requirements. The torpedo control cable shall withstand the following environmental conditions without damage, and shall remain in conformance with all other requirements specified herein.

3.8.1 High temperature. The assembly shall withstand exposure to 130° ± 5°F (54° ± 3°C) for 48 hours when tested in accordance with 4.6.9.

3.8.2 Low temperature. The cable assembly shall withstand exposure to -20° ± 5°F (-29° ± 3°C) for 48 hours when tested in accordance with 4.6.10.

3.8.3 Shock. The cable assembly shall meet the requirements of 3.6.2, 3.7.2, 3.7.3, and 3.7.4 after being subjected to shock tests in accordance with 4.6.11.

3.8.4 Vibration. The cable assembly shall meet the requirements of 3.6.2, 3.7.2, 3.7.3, and 3.7.4 and there shall be no circuit discontinuities

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exceeding 100 microseconds in duration after the cable assembly has been subjected to vibration tests in accordance with MIL-STD-167-1, Type I, and Section 4.6.12 of this specification.

3.8.5 Hydrostatic pressure. The cable assembly shall withstand exposure to the hydrostatic pressure specified in Test No. 8 of Dwg. 821024 without showing evidence of electrical breakdown or water leakage when tested in accordance with 4.6.13.

3.8.6 Marine environment. The cable assembly shall withstand exposure to the sea atmosphere and salt water immersion conditions specified in 4.6.14.

3.8.7 Temperature and humidity cycling. The cable assembly shall withstand exposure to two 14-day temperature and humidity cycles as specified in 4.6.15.

3.8.8 Tropical environment. The assembly shall withstand exposure to the tropical environment conditions specified in 4.6.16.

3.9 Workmanship. The cable assemblies shall be constructed and finished in a manner that will ensure compliance with all requirements of this specification and applicable NAVSEA drawings. Particular attention shall be paid to the quality of rubber molding, plastic molding, and finishes, and to freedom from defects, mold flash, and burrs or sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of workmanship insofar as not specifically covered by applicable specifications. See 6.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. See 6.2.

4.2 Classification of inspections. Inspection of the control cable assembly shall be classified as follows:

a. Preproduction inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

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4.3 Preproduction inspection.

4.3.1 Preproduction inspection procedure. The preproduction sample of four cable assemblies, together with the mating adapters and connectors listed in table I, shall be subjected to the tests and examinations specified for preproduction samples in table II. The test sequence shall be as specified in table II.

TABLE I. Preproduction samples and mating connectors.

Quantity	Description	NAVSEA Number	Sample Number
4	Cable Assemblies	LD 162420, Dwg 783552 (preproduction samples)	1C-4C
4	Torpedo Connector Receptacles	LD 162596, Dwg 758008, or PL 2865969	1A-4A
As req'd	Shear Screws	Dwg 783585	N/A
1	Torpedo Tube Cable Connector	LD 162838, Dwg 2865440	M-1

4.3.2 Mating connectors. Mating connectors, when required, will be furnished by the procuring activity. See 6.2.

4.3.3 Inspection data. When specified in the contract or purchase order, certified inspection data covering the results of preproduction inspections shall be forwarded to the procuring activity. See 6.2.

4.3.4 Defects. A preproduction sample cable assembly which fails to conform to any requirement verified by test or examinations specified in table II shall be cause for denial of production approval. See 6.2. Further production of cable assemblies by the supplier prior to the approval of the preproduction sample shall be at the supplier's risk.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following:

a. Individual inspection

b. Periodic production inspection

4.4.1 Individual inspection. Each cable assembly shall be subjected to the examinations and tests specified in table III (100 percent inspection).

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TABLE II. Preproduction and periodic production inspection.

Examination or Test	Requirement Paragraph	Method Paragraph	Preproduction Samples Only		Periodic Production Sample Numbers	
			Sample Numbers	Mating Connector Numbers	First Lot	Succeeding Lots
Examination of Product	3.4,3.5,3.9	4.6.1	1C-4C		1-6	1-4
Quality Conformance Test			1C-4C	1A-4A	1-6	1-4
Connector Mating and Unmating	3.6.2	4.6.3	1C-4C	1A-4A, M-1	1-6	1-4
High Temperature	3.8.1	4.6.9	1C, 2C	1A, 2A	3-5	1,2
Low Temperature	3.8.2	4.6.10	3C, 4C	3A, 4A	3-5	1,2
Shock	3.8.3	4.6.11	1C-4C	1A-4A	3-5	1,2
Vibration	3.8.4	4.6.12	1C-4C	1A-4A	3-5	1,2
Cable Pull Test	3.6.1	4.6.2	1C-4C	1A-4A, M-1	1-6	1-4
Electrical Requirements	3.7	4.6.4-4.6.7	1C-4C	1A-4A	1-6	1-4
Marine Environment	3.8.6	4.6.14	3C	M-1, 3A	6	1,2
Temperature and Humidity Cycling	3.8.7	4.6.15	1C, 2C, 3C	1A, 2A, 3A, M-1	1	3
Tropical Environment	3.8.8	4.6.16	4C	4A, M-1	2	4
Hydrostatic Pressure	3.8.5	4.6.13	1C-4C	1A-4A M-1	1-6	1-4

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4.4.1.1 Failure of a cable assembly to conform to any requirement verified by the examinations or tests specified in table III shall be cause to reject that cable assembly as defective. See 6.2.

TABLE III. Individual inspection.

Examination or Test	Requirement Paragraph	Method Paragraph
Examination of Product	3.4, 3.5, 3.9	4.6.1
Dielectric Strength	3.7.2	4.6.5
Insulation Resistance	3.7.3	4.6.6
Cable Continuity	3.7.5	4.6.8
Hydrostatic Pressure	3.8.5	4.6.13
Packaging and Packing	5	4.6.17

4.4.2 Periodic production inspection.

4.4.2.1 Lot. A "lot" shall consist of a continuous run of cable assemblies produced by the same process and with the same equipment over a period not longer than three months and in a quantity not greater than 150 units.

4.4.2.2 Sampling plan. Periodic production samples shall be selected from those cable assemblies which have passed the individual inspection specified in 4.4.1 herein, as follows:

a. Step 1. Upon the initiation or resumption of production, randomly select six units from each lot of 150 units. Upon satisfactory acceptance of samples selected from the first lot, shift to Step 2.

b. Step 2. Randomly select four units from each lot of 150 units. Continue with step 2 for the balance of the contract or until a defective unit is found or until a change in manufacturing methods or materials is introduced, in which event sampling shall revert to step 1.

4.4.2.3 Periodic production inspection procedure. The periodic production samples shall be subjected to the tests and examinations specified in table II in the sequence indicated.

4.4.2.4 Failures. A periodic production sample failing to conform to any requirement verified by the examinations or tests specified in table II shall constitute a defect and shall be cause for rejection of the lot represented by the sample and may be cause for revocation of production approval. See 6.2.

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4.5 Test conditions. Unless otherwise specified, the tests and examinations shall be conducted under any combination of conditions within the ranges below.

a.	Temperature:	20° to 30°C.
b.	Relative humidity:	Room ambient to 95 percent.
c.	Barometric pressures:	24 to 31 inches of mercury. (80 to 105 kilopascals (kPa))

4.6 Inspection procedures.

4.6.1 Examination of product. The cable assemblies and components shall be examined to ensure conformance to the applicable NAVSEA drawings and to the requirements in (a) and (b) below. Assurance shall be made that no changes have been made in the cable assemblies other than those approved prior to submission of the preproduction sample:

a. Materials, design, and construction

b. Workmanship

4.6.2 Torpedo cable assembly pull test. The tube door plug and torpedo plug of the cable assembly shall be mounted as shown in figure 1 and a tensile force of 300 pounds + 10, - 0 pounds (1335 + 45, - 0 N) shall be exerted on the cable for a period of 30 seconds minimum along the longitudinal axis of the connector's cable entrances. During this test, each of the 65 cable circuits shall be electrically checked for continuity with a 0-5 Vdc buzzer or equivalent. Discontinuous circuits, as indicated by failure of the buzzer to sound, shall indicate a failure.

NOTE: The continuity check shall be started immediately after application of the 300-pound (1335-Newton) force and shall be completed as soon as possible thereafter. The tensile force shall be applied no longer than is necessary to perform the continuity check.

4.6.3 Connector mating and unmating force. Securely mount a Torpedo Tube cable Connector, LD 162838, in a movable holding device as shown in figure 2. The holding device shall be capable of measuring force in both a push and pull direction. The tube door plug shall be engaged to and disengaged from the mating connector by hand as in service use. A force required to engage and disengage the plug greater than 50 pounds (222 N) shall indicate a failure.

4.6.4 Current carrying capacity. A mated cable assembly and torpedo receptacle shall be used for this test. Connect circuits F, AA, and AB in series and, utilizing a suitable power supply, energize this circuit with 5.5 $\pm$ 0.1A at 400 Hz for 20 minutes. The current shall then be

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increased to 9.2 A for 3 + 1, - 0 seconds. Next connect circuits R, h, and AG in series and energize the circuit with 5.5 ± 0.1 A at 60 Hz for 20 +1, - 0 minutes. The current shall then be increased to 9.2 ± 0.2 A for 3 + 1, - 0 seconds. At the conclusion of this test the mated test specimen shall be subjected to the dielectric strength test and circuit resistance test. The test specimen shall then be unmated and visually examined for any physical degradation.

4.6.5 Dielectric strength. Apply a 725 ± 25 volts dc or 515 ± 15 volts ac potential between two points, one being each circuit in turn and the other being all remaining circuits and the shell connected together to ground. The assembly shall be examined for evidence of current leakage or dielectric breakdown, as indicated by arcing. This test may be performed with the test specimen in the mated condition.

4.6.6 Insulation resistance. Apply a 500-750 Vdc potential between two points, one being each circuit in turn and the other being all the remaining circuits and the shell connected together to ground. The potential shall be applied until a stable reading is attained, or for two minutes, maximum. The mated test specimen insulation resistance shall be determined. Mated resistance less than one megohm shall indicate failure. The insulation resistance of the unmated test specimens shall be determined. Unmated insulation resistance less than 10 megohms shall indicate failure.

4.6.7 Circuit resistance. With the cable assembly and Torpedo Connector Receptacle, LD 162596, mated, measure the resistance of each of the 65 circuits, using a Kelvin Bridge or equivalent (see figure 3). The potential across the circuit shall be kept to a minimum and shall not exceed 5 volts. A circuit resistance exceeding 0.50 ohm shall indicate failure.

4.6.8 Torpedo cable continuity. Each circuit shall be probed for continuity with a 0-5 Vdc buzzer or equivalent. Evidence of discontinuity shall indicate failure.

4.6.9 High temperature. The mated cable assembly and torpedo connector receptacle shall be mounted in a test chamber. The chamber temperature shall be raised to 130° ± 5°F (54° ± 3°C) and sustained for a period of not less than 48 hours. At the end of this period, the test specimen, while still at the test temperature of 130°F (54°C), shall be subjected to the insulation resistance and circuit resistance tests.

4.6.10 Low temperature. The mated cable assembly and torpedo connector receptacle shall be mounted in a test chamber. The chamber temperature shall be lowered to -20° ± 5°F (-29° ± 3°C) and maintained at this temperature for a period of not less than 48 hours. At the end of this time, and while still at the test temperature, the test specimen shall be subjected to the insulation resistance and circuit resistance tests.

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4.6.11 Shock. The mated cable assembly (see figure 4) shall be subjected to shock tests in accordance with MIL-S-901C for Grade A, Class I, Lightweight Equipment after which the test specimen shall be subjected to 4.6.3, 4.6.5, 4.6.6, and 4.6.7.

4.6.12 Vibration.

4.6.12.1 Vibration (transportation). The cable assembly, packaged in its individual shipping container, shall be secured to the vibration table such that the means of securing will not deform the shipping container. The cable assembly in its shipping container shall be vibrated in each of its three mutually perpendicular axes. The vibratory input level shall be maintained at or near the shipping container area of contact with the vibration table. If the input is measured at more than one control point the control signal shall be the average of all input accelerometers. Sinusoidal cycling shall be conducted at a logarithmic rate for 84 minutes per axis in accordance with Schedule I.

SCHEDULE I

Frequency Range	Amplitude	Sweep Time
5-200HZ Tolerance: +0.5HZ@<25HZ ± 2%@≥25HZ	1.5gpk ± 0.15	5-200-5HZ 12 minutes

4.6.12.2 Vibration (ready, nonoperating). The mated cable assembly and torpedo connector receptacle shall be mounted in a vibration fixture (see figure 4). The test fixture shall be rigidly attached to the vibration table. All circuits of each mating connector shall be wired in series for this test. The vibratory input control accelerometer shall be located on the vibration fixture adjacent to the cable tube door mounting point. Two monitoring accelerometers shall be mounted on the cable assembly, one located on the torpedo end and one on the tube door end. The test item shall then be subjected to vibration in each of its three mutually perpendicular axes. Sinusoidal cycling shall be conducted in accordance with Schedule II. When conducting the exploratory vibration test the change in frequency shall be made in discrete frequency increments of 1 HZ and maintained at each frequency for not less then 15 seconds. The frequency at which resonances occur shall be noted. During the variable frequency test the change in frequency of vibration shall be made in discrete frequency increments of 1 HZ and maintained at each frequency for 5 minutes. The series circuit shall be monitored during vibration for discontinuity, using a relay chatter indicator or equivalent. The circuit discontinuity shall not exceed 100 microseconds.

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SCHEDULE II

Type of Test	Frequency (HZ)	Table Amplitude (Inches - peak to peak)
Exploratory	4-33	0.020 ± .004
	34-50	0.006 ± 0.000 - 0.001
Variable Frequency	4-15	0.060 ± 0.012
	16-25	0.040 ± 0.008
	26-33	0.020 ± 0.004
	34-40	0.010 ± 0.002
	41-50	0.006 ± 0.000 - 0.001
Endurance	At the resonant frequencies determined from variable frequency test.	At amplitude corresponding to frequency selected for endurance test.

The endurance test of Schedule II shall be conducted by vibrating the test item for a total period of at least 2 hours at the resonant frequencies chosen by the test engineer based on the variable frequency vibration sequence. If no resonant condition results the test shall be performed at 50 Hz. After these vibration tests the control cable shall be subjected to and successfully complete the tests specified in 4.6.3, 4.6.5, 4.6.6, and 4.6.7.

4.6.13 Hydrostatic pressure. The cable assembly shall be mounted in a hydrostatic pressure tank simulating, as in use, the tube door and torpedo connector receptacles (see figure 5). Subject the mated units to not less than 5 psi (34.5 kPa) external hydrostatic pressure. Hold under this pressure for not less than 15 minutes. Examine for leakage. Leakage shall indicate failure. During this period, perform the dielectric strength, insulation resistance, and continuity tests. Upon completion of the electrical tests, completely relieve the pressure. Subject the mated units to Test No. 8 of Drawing 821024. During this period perform the dielectric strength, insulation resistance, and continuity tests. Upon completion of the electrical tests, separate the cable assembly from the torpedo connector receptacle. When subjected to Test No. 8 of Drawing 821024, cable assembly separation from the torpedo connector receptacle with an axial force exceeding 300 pounds shall indicate failure. Upon completion of the test, remove the cable assembly and torpedo connector receptacle from the pressure tank, thoroughly dry the mating surfaces, and perform the current capacity, dielectric strength, insulation resistance and circuit resistance tests. (NOTE: Perform cable separation test only on preproduction and periodic production samples.)

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4.6.14 Marine environment.

4.6.14.1 Sea Atmosphere. The unmated cable assembly and torpedo adapter, ready for service use, shall be exposed for a period of not less than 96 hours to sea atmosphere as experienced at dockside or on the main weather deck of a ship. The test specimens shall be protected from other environmental conditions in such a manner as to permit free circulation of air around the specimens.

4.6.14.2 Immersion. Upon completion of the sea atmosphere test, the cable assembly shall be mated to connectors as in service use. The complete test specimen shall then be immersed in natural sea water for not less than 4 hours and removed for a period of 30 + 2 minutes. (The rears of the mating connectors shall not be immersed. They shall be protected in such a manner so as not to be exposed to the salt water.) This cycle shall be repeated eight times. No fresh water rinse is permitted during these cycles. With mating connectors removed, the cable assembly and torpedo adapter shall then be subjected to the following tests in the sequence shown:

NOTE: The specimens shall not be dried, baked, or wiped prior to performing these tests:

a. Insulation resistance

b. Circuit resistance

c. Mating and unmating force

4.6.15 Temperature and humidity cycling. The cable assembly and torpedo connector receptacle shall be mated for this test. The coupled unit shall be mounted in the chamber wall or suitably protected so as to prevent exposing the tube door connector face or torpedo receptacle rear to moisture. The total test is made up of two 14-day temperature and humidity cycles. As used in this test, the term "chamber temperature" is the temperature of the air immediately surrounding the test specimen. Chamber temperature may be changed (1) by varying the temperature of a single chamber, or (2) by moving the test specimen from one constant temperature chamber to another. The fluctuations from all specified temperatures shall not exceed 10°F (6°C). There shall be no deviation from the test sequence as prescribed.

4.6.15.1 Place the test specimen in a chamber which has been maintained at -65°F (-54°C) for at least 2 hours immediately prior to start of test (humidity not controlled). After 8 hours exposure, the chamber temperature shall be changed to +160°F (71°C) and 95 percent humidity in less than 2 hours. The test specimen shall be exposed to these conditions for not less than 14 hours, after which time the temperature shall be decreased. The rate of decrease of chamber temperature shall be equal to or greater than 36°F (20°C) per hour for at least 2.5 hours. The temperature of the

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air surrounding the test specimen shall be +70°F (21°C) or lower. The chamber temperature shall reach -65°F (-54°C) in not more than 3 1/2 hours. This temperature shall be held for not less than 2 hours. The chamber temperature shall then be increased to +160°F (71°C) and 95 + 3, - 0 percent humidity as rapidly as practicable and maintained for not less than 16 hours. Three of the above temperature cycles, starting with temperature decrease, shall be performed.

4.6.15.2 At the conclusion of the temperature cycles as outlined above, the chamber temperature shall be increased to +160°F (71°C) and 95 percent humidity and maintained for not less than 64 hours. The sequence of operations, starting with temperature decrease, shall be repeated every 24 hours for 5 days.

4.6.15.3 The chamber temperature shall be reduced to -65°F (-54°C) and maintained for not less than 16 hours. This sequence of temperature and humidity conditions constitutes one cycle. Two such cycles shall be performed. At the conclusion of the second cycle, the test specimen shall be allowed to return to room temperature and shall be subjected to the dielectric strength, insulation resistance, and circuit resistance tests.

NOTE: Insulation resistance of the test specimen shall be measured during the temperature and humidity cycle, 20 percent of the circuits being selected at random for insulation resistance monitoring. The insulation resistance shall be monitored after the first 24 hours of testing and every 168 ± 1 hours thereafter.

4.6.16 Tropical environment. The mated cable assembly and torpedo connector receptacle shall be mounted in the chamber wall or suitably protected so as to prevent exposing the tube door connector face or torpedo receptacle rear face to fungi. The test specimen shall be placed in a mold chamber in which a temperature of +86° ± 5°F (30° ± 3°C) and relative humidity of 90 percent ± 5 percent is maintained. The test specimen shall be sprayed with a suspension of mixed spores. The test specimen shall be maintained in this condition for not less than 28 days. At the conclusion of this period, the test specimen shall be returned to room temperature and examined to determine if it is a nutrient to fungi. If the test specimen shows susceptibility to fungi, colored photographs shall be taken. At least five groups of fungi shall be used in test specimen inoculations. Five such groups are listed in MIL-T-18404 and one type of fungus from each group shall be used. In preparing the spore suspension, distilled water (having a pH value between 5.8 and 7.2 and a temperature of 77°F (25°C) shall be prepared by sterilization in convenient containers (approximately 100 milliliters each). Approximately 10 milliliters of the solution shall then be introduced directly into a stock culture and shaken vigorously in order that a well-sporulated suspension will result without distrubing the agar. This process shall be repeated for each type fungus. The separate spore suspensions from the various types of fungi shall then be mixed together in an atomizer to

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provide a composite suspension. Actively growing cultures in ripe fruiting condition shall be used for each inoculation and under no conditions will the inoculum be kept for more than an 8-hour period. Following its exposure to this environment, the test specimen shall be removed from the chamber and shall be subjected to the dielectric strength, insulation resistance, circuit resistance, and cable mating and unmating force tests.

4.6.17 Packaging and packing. The packaging and packing of the cable assembly shall be examined to detemine conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The cable assembly shall be packaged in accordance with Method III of MIL-P-116 and LD 162422.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The cable assembly shall be packaged in accordance with manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Consolidation container. The cable assembly, packaged as described in 5.1.1.1, shall be packed in a fiberboard box in accordance with PPP-B-636 and LD 162422 and may be shipped individually in small lots.

5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 100. Battens shall be utilized to eliminate the underhang. See 6.2.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The cable assembly, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for

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immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129. See 6.2.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. Torpedo Control Cable Mk 6 Mod 0 is intended for use aboard submarines and surface vessels in transmitting fire control electrical signals from the torpedo tube door to Torpedo Mk 48 and Mobile Target Mk 27. After firing, the assembly can be cleaned, inspected, and reused in additional firings.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements

a. Title, number, and date of this specification.

b. Any exceptions to this specification.

c. Activity at which inspections will be performed. (4.1)

d. Number of preproduction samples, if other than four. (3.3)

e. Materials or parts to be used other than those listed in LD 162420. (3.4)

f. Requirements for and disposition of a preproduction sample to be used to define standards of workmanship. (3.9).

g. Number and type of mating connectors to be furnished by the government, and the circumstances in which they will be furnished. (4.3.2).

h. Disposition of defective and failed items. (4.3.4, 4.4.1.1, 4.4.2.4)

i. Levels of preservation and packaging. (5.1)

j. Level of packing. (5.2)

k. Palletizing details other than those covered in MIL-STD-147. (5.2.1.2)

1. Markings other than those in accordance with MIL-STD-129. (5.3)

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6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID
4.3.3	Certified inspection data	DI-T-2072

These data will be delivered as identified on the applicable DID only when specified on DD Form 1423 (Contract Data Requirements List) and incorporated into applicable contracts.

(Copies of data item descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Metrication. This document has been annotated parenthetically with metric "equivalents" of quantities not expressed in the International System of Units (SI). Metric units are in accordance with ASTM E380-72 as adopted by the DoD on 29 November 1973, and are included because the DoD encourages conversion to metric dimensioning. Parenthetical quantities are intended as information only, as they are usually not exact equivalents, but rounded off within specified tolerances.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Navy - OS

Preparing Activity
Navy - OS
Project Number:
1355-N392

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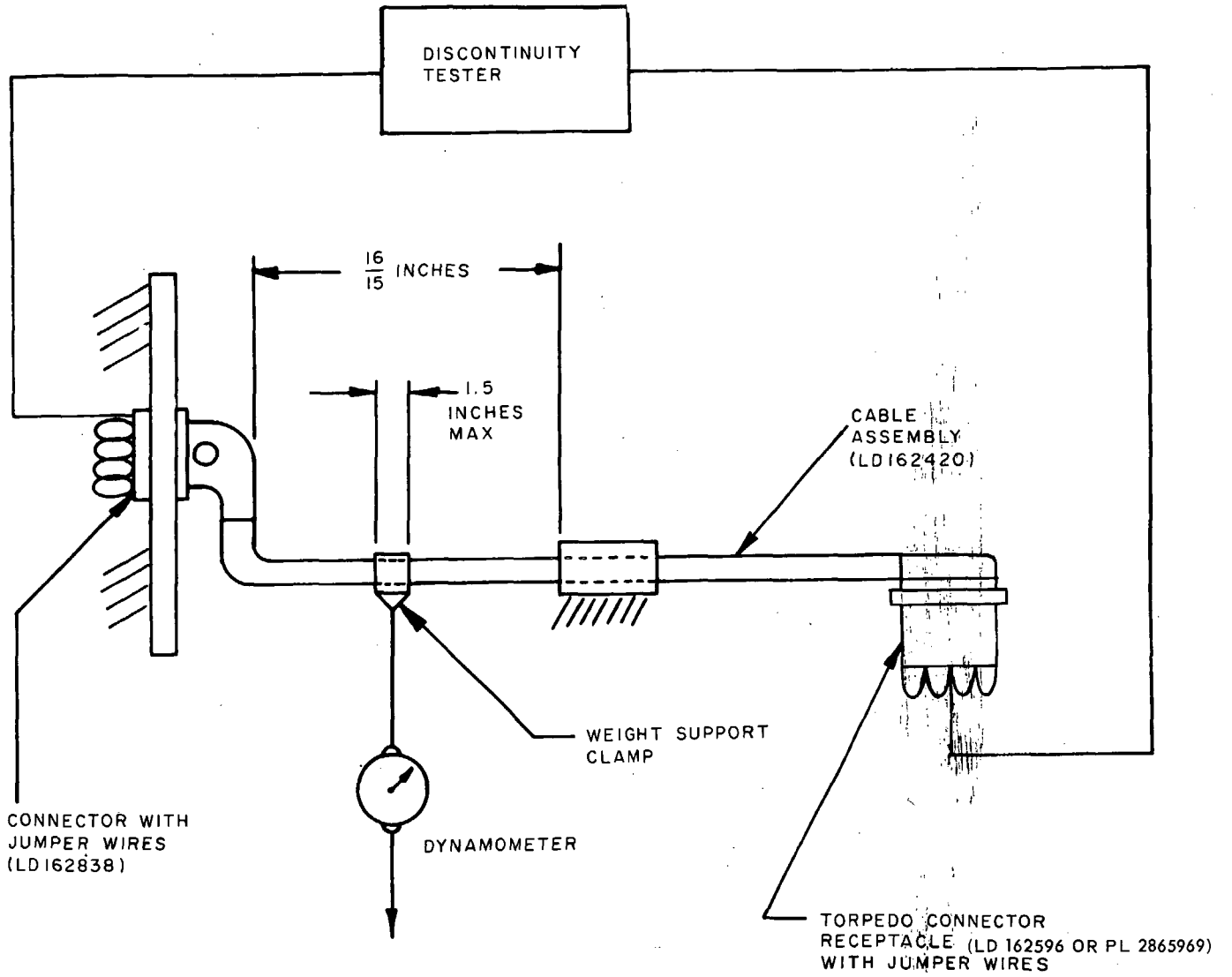


FIGURE 1. Test Setup - Cable Pull Test.

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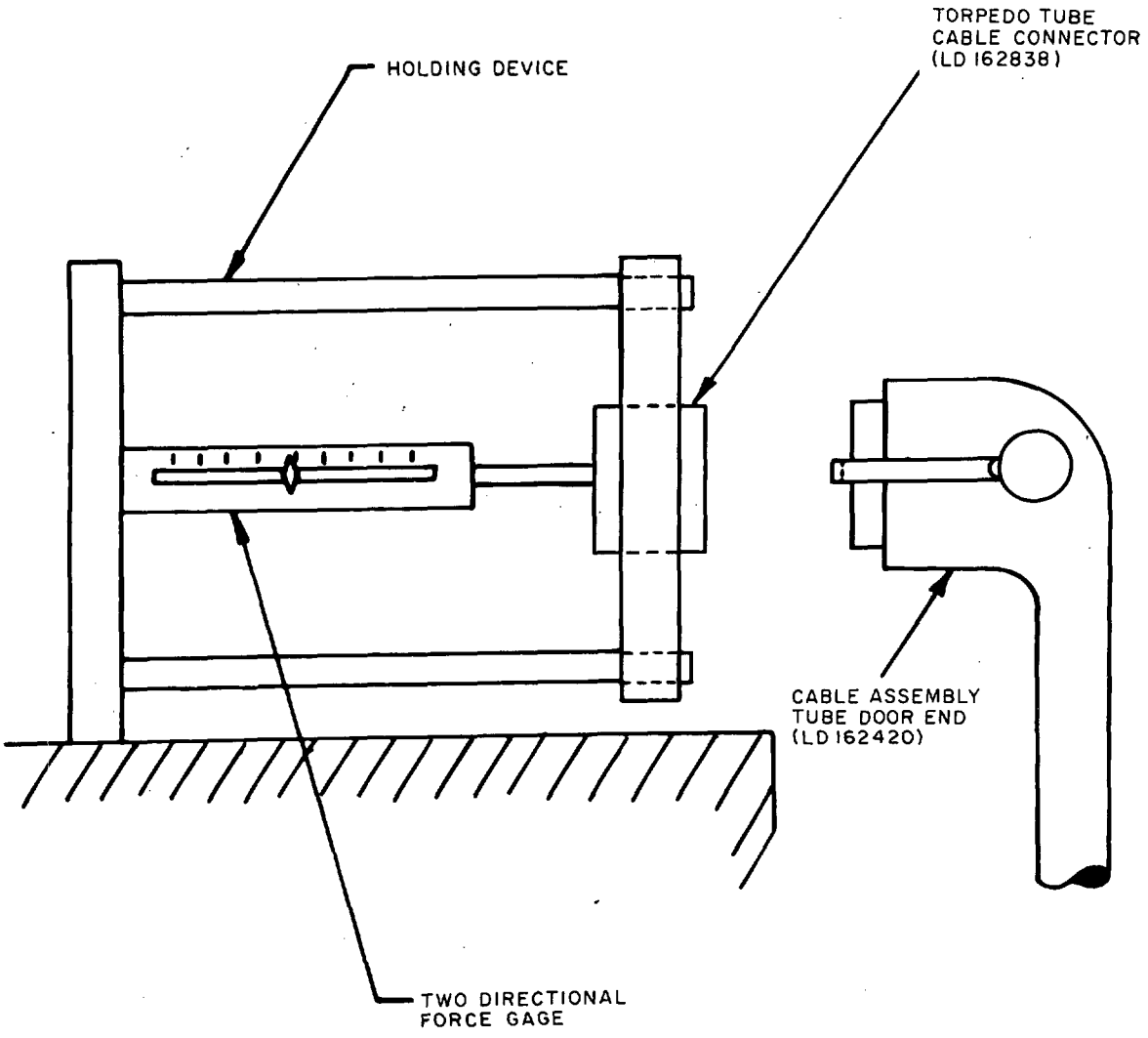


FIGURE 2. - Test Setup - Mating and Unmating Force.

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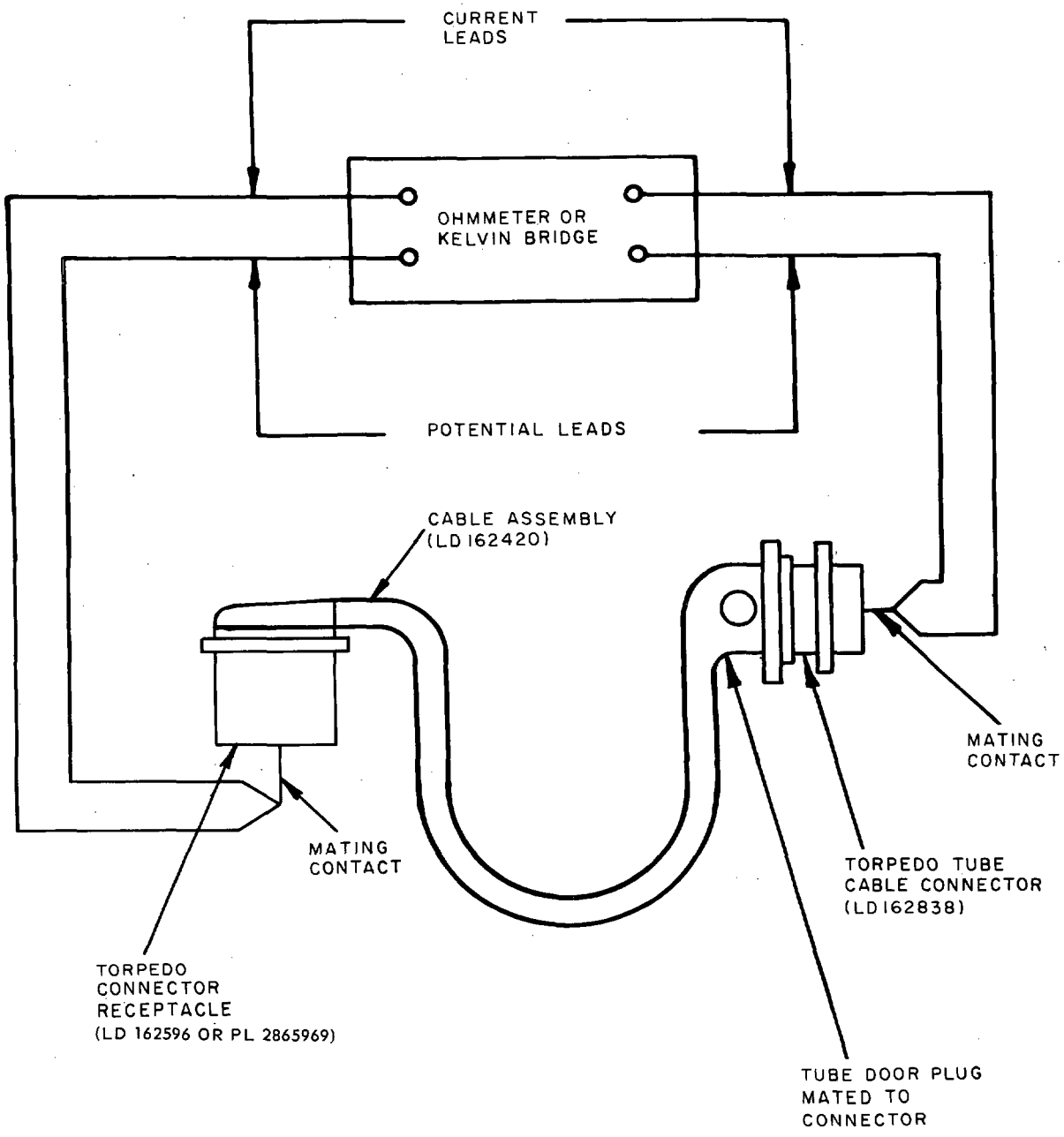


FIGURE 3. Test Setup - Circuit Resistance.

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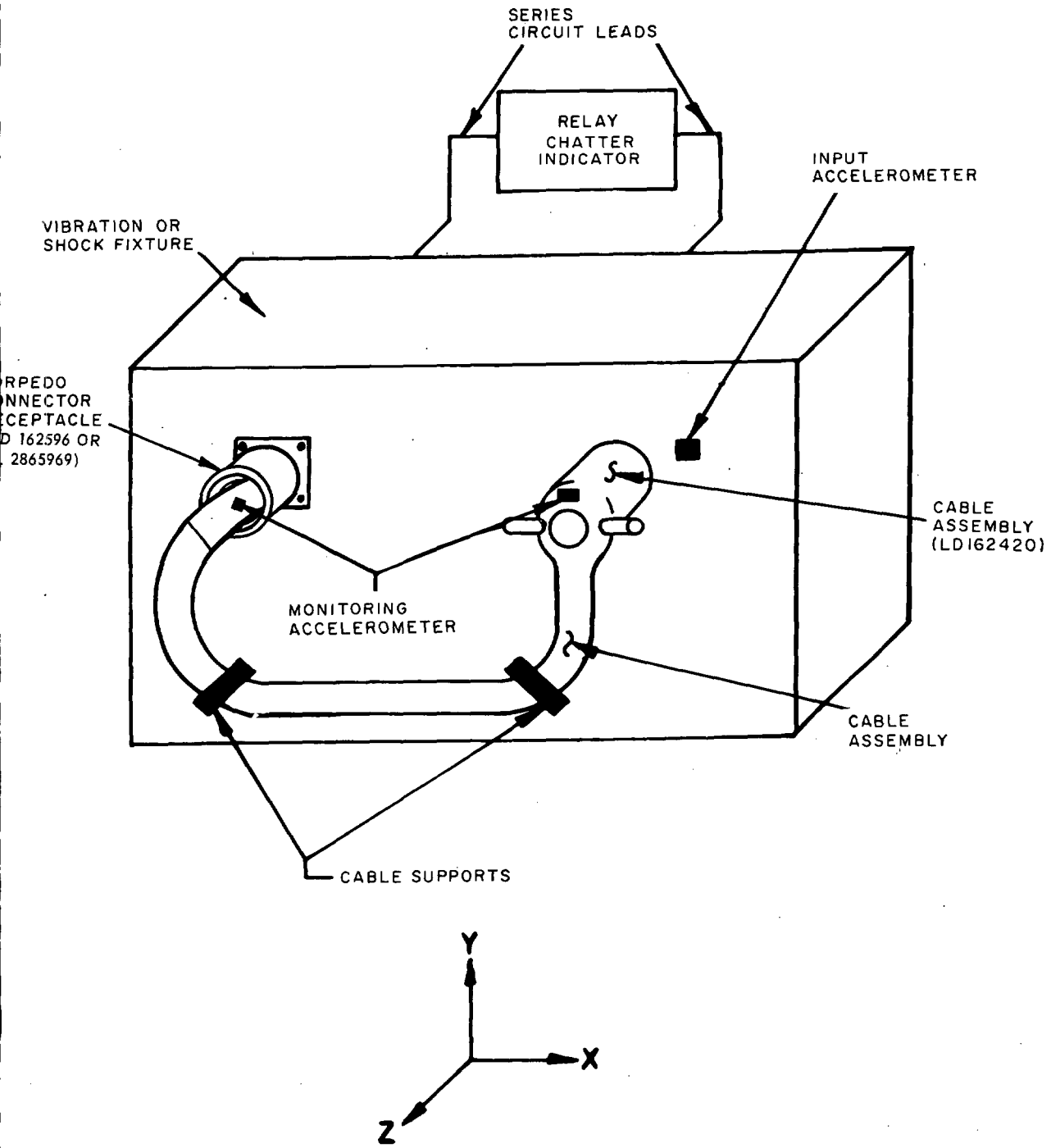


FIGURE 4. Test Setup - Vibration and Shock.

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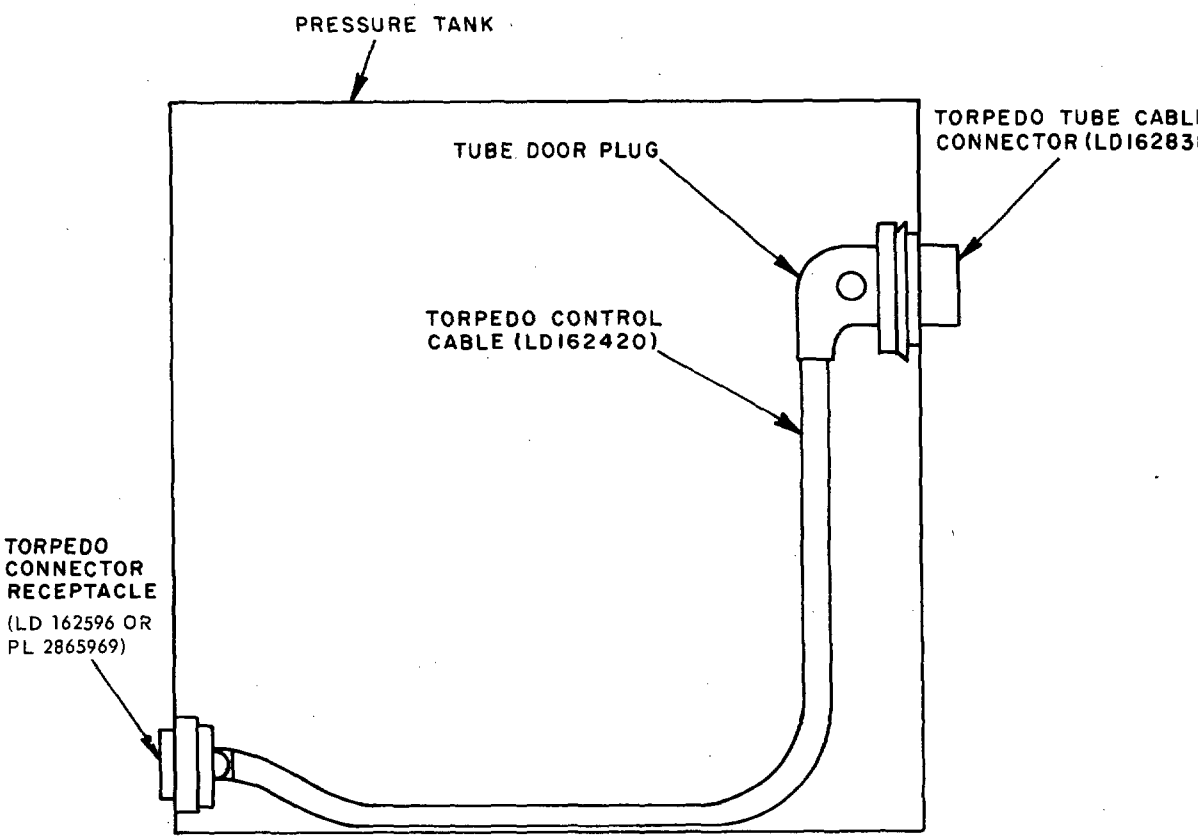


FIGURE 5. Typical Test Setup - Hydrostatic Pressure.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-82614C (OS) CABLE ASSEMBLY, TORPEDO CONTROL, MARK 6 MOD 0

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-LF-014-1802

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